

● AI INFRA SUMMIT HACKATHON 2026

Intel Track: Bimanual VLA Manipulation

Speechmatics Best-Use Bonus

PROJECT SUBMISSION • DETERMINISTIC SAFETY LAYER

BOLLARD.

The rigid gate between hearing and hands: **nothing moves** until the voice command passes deterministic policy and cryptographic audit.

• Sub-200ms Speechmatics Real-time

• Deterministic Grammar (Zero-LLM in loop)

• Intel Bimanual Dual-Arm Kinematics

• Cryptographic Hash Chain Audit

Author: Raphael Ehindero • Live: <https://trybollard.netlify.app> • Code: github.com/A-Raphie/bollard

Voice to Robot Execution is Catastrophic Without Physical Guardrails

VLA (Vision-Language-Action) models and LLMs hallucinate coordinates, misunderstand spoken homophones, and ignore physical envelope limits when executing dual-arm manipulation.

FAILURE MODE 01

Acoustic Misparsing

In noisy industrial environments, speech engines drop negatives, confuse homophones ("right" vs "light"), and mangle units ("10 cm" vs "10 meters"), leading directly to high-velocity collisions.

FAILURE MODE 02

Physics & Envelope Blindness

LLMs lack real-time kinematic spatial awareness. They happily issue commands exceeding dual-arm reach, collision paths between left and right end-effectors, or over-payload grasp limits.

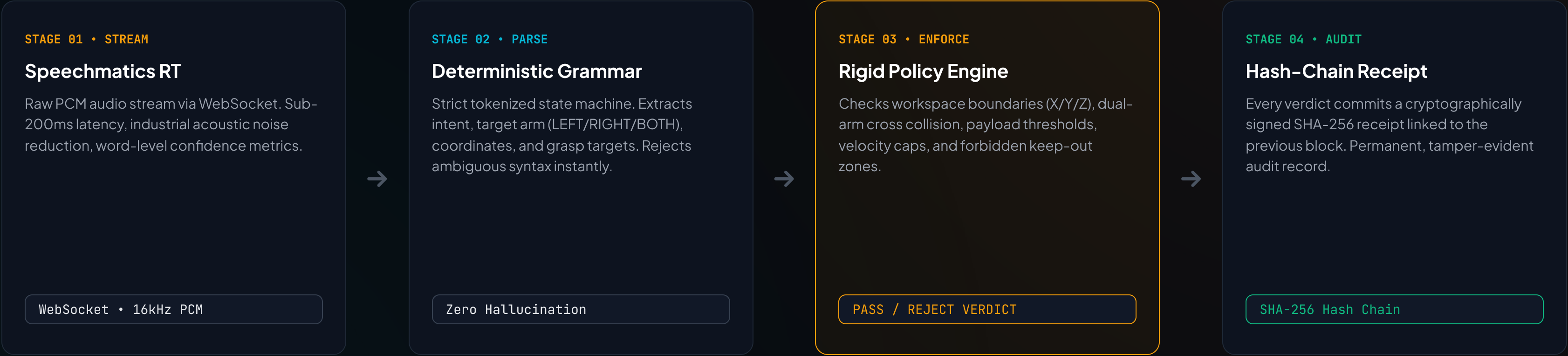
FAILURE MODE 03

Zero Audit Trail

When an industrial arm crushes hardware, post-incident investigations find ambiguous logs. There is no verifiable proof of what audio was transcribed, what rule passed, or what actuator moved.

The Deterministic Pipeline: Zero LLMs in the Critical Path

Every spoken word flows through 4 uncompromising, mathematical stages before reaching robot actuators.



Engineered for Bimanual Robotic Arms

Addressing the specific multi-arm coordination challenges of the Intel Bimanual VLA manipulation track.

BIMANUAL SAFETY KINEMATICS

Dual-Arm Collision & Handover Zones

- **Independent Arm Addressing:** Direct voice steering for Left Arm, Right Arm, or Synchronized dual-arm tasks.
- **Mutual Exclusion Spheres:** Prevents end-effectors from occupying overlapping coordinate voxels during uncoordinated motions.
- **Handover Protocols:** Enforces bilateral hold state confirmation before the releasing arm disengages its gripper.

OPERATOR TELEMETRY & SIMULATION

Real-Time Interactive 2D/3D Cockpit

- **Live 60 FPS Canvas:** Renders dual articulated arms, target workpieces, and danger zones in real-time.
- **Immediate Visual Feedback:** Visualizes green approved trajectory vectors vs red blocked collision barriers instantly.
- **Direct Manual Overrides:** Hardware E-STOP and one-click emergency halts integrated directly into the operator HUD.

Cryptographic Receipts: Tamper–Evident Audit Ledger

Replacing opaque robot log files with mathematical proof of policy compliance.

01 • CANONICAL PAYLOAD

Deterministic Serialization

Every command record serializes:

```
{ id, timestamp, transcript, arm: "BOTH", rule:
  "POLICY_ENVELOPE", verdict: "ALLOWED", prevHash }
```

02 • HASH LINKING

SHA–256 Merkle Chain

Each receipt embeds the cryptographic hash of the preceding receipt. Any modification to past voice commands or verdicts immediately breaks the chain validation.

03 • INSTANT VERIFICATION

Client–Side Proof

Operators and regulatory inspectors can verify the entire sequence with zero reliance on cloud trust. 100% verifiable offline and on-premise.

Bollard is Shipped, Tested, and Live

A production-grade voice safety boundary for the next generation of bimanual robotics.



Live Production App

trybollard.netlify.app

Fully responsive landing page & interactive cockpit



Interactive Cockpit

trybollard.netlify.app/app

Real-time Speechmatics mic, canvas sim & receipts



Open Source Repo

github.com/A-Raphie/bollard

Clean TypeScript architecture with full test suites

"Nothing moves until the command passes policy."